

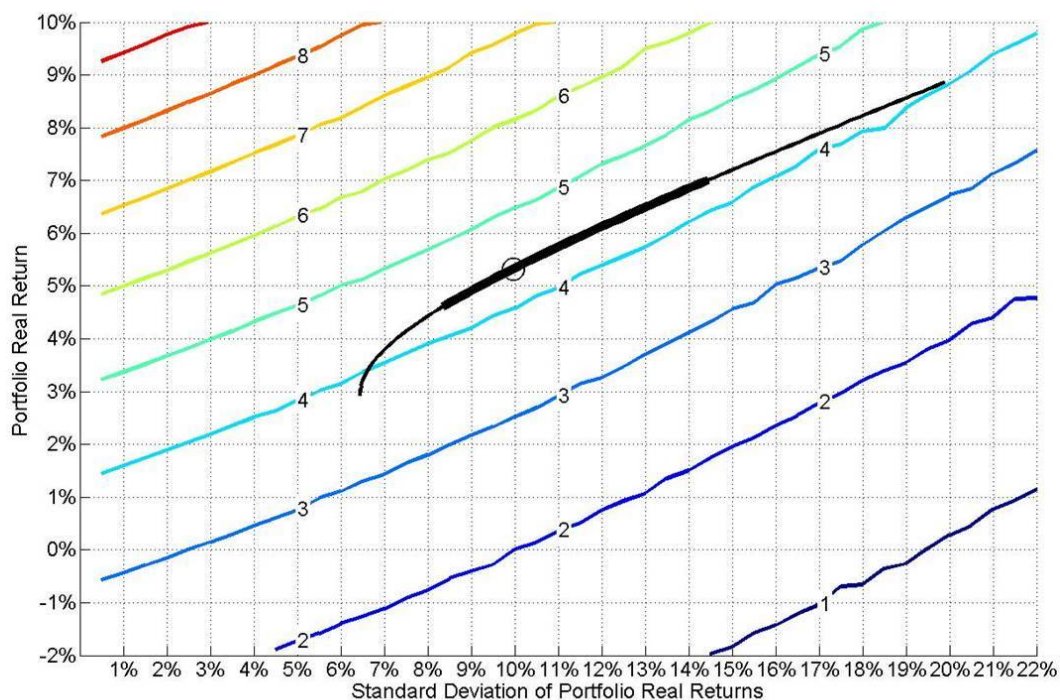


Exhibit 4.7 shows how sustainable withdrawal rates relate to the expectations about portfolio returns and volatilities when the allowed failure rate is set at 10 percent and the retirement horizon is set at thirty years. As we move toward the upper left-hand corner of the exhibit, higher sustainable withdrawal rates are supported through combinations of higher portfolio returns and lower portfolio volatilities. We can observe the lines in the exhibit that map out different withdrawal rates and how returns and standard deviations offset each other in order to maintain the same withdrawal rate.

The efficient frontier of asset allocations generated from the historical data, when combining large-capitalization stocks (S&P 500) with intermediate-term government bonds, is included in this exhibit. The points on this curve represent the highest possible returns for a given level of volatility, or the lowest volatility achievable with a given return. It is the standard formulation for an efficient frontier used in modern portfolio theory, but it is overlaid on a graph that provides information about how different points on the frontier impact sustainable spending rates.